Discrete Drainage Dynamics IV: Light Deflection, Shapiro Delay, and the Photon-Substrate Coupling in the Static Spherical Sector

A phenomenological calibration of the spatial bandwidth coupling

Stanislas Dewavrin[1]

[1]Independent researcher , dewavrin.iphone@gmail.com

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Abstract

Discrete Drainage Dynamics (DDD) is an analogue-substrate programme. It does not contest General Relativity: GR is the empirical anchor any analogue must reproduce in the appropriate weak-field, static spherical limit. We ask the exploratory question of whether two canonical weak-field GR observables — light deflection and Shapiro delay — can emerge from the discrete toy substrate of Papers II and III under a phenomenological photon-substrate coupling.

Paper II derived a static spherical bandwidth profile $R(r)/R_0 = \sqrt{1-\beta/r}$ and identified its time component with a Schwarzschild-like clock-rate factor. Paper III introduced the spinor sector and recovered the Lorentz kinematic factor at moving wavepackets. Here we ask the next question: does DDD reproduce the gravitational propagation observables — light deflection $4GM/c^2b$ and the Shapiro time delay — in the same static spherical sector? We do not derive the photon sector from the substrate dynamics; that is deferred to Paper VI. We adopt a phenomenological coupling, $c_{\text{eff}}(r) = c\,(R/R_0)^\eta$, and test two ansatze: $\eta = 1$ (temporal only) and $\eta = 2$ (temporal + spatial). Numerical ray-tracing through the Paper II profile gives, for $\eta = 2$, asymptotic convergence to the GR predictions $\Delta\theta = 4GM/c^2b$ (deflection) and $c\,\Delta T = (4GM/c^2)\ln(\cdots)$ (Shapiro); in our numerical setup the Shapiro delay reaches sub-percent agreement near $b/\beta \sim 50$, the deflection near $b/\beta \sim 500$, with finite-impact-parameter corrections behaving as $\sim 2.4\,\beta/b$ in both observables. The $\eta = 1$ ansatz gives exactly half the GR values, ruled out by data at the factor-of-two level. The choice $\eta = 2$ is the substrate analogue of the spatial-component contribution of the Schwarzschild metric; $\eta = 1$ would be the prediction if the spatial part of the substrate were rigid. Within this two-member ansatz family the data falsify $\eta = 1$ and select $\eta = 2$. Following the hypothesis terminology established in Paper III, we adopt $\eta = 2$ as a structural *hypothesis* of the photon-substrate coupling, not as a derived consequence: the substrate principles fix the temporal component $g_{tt} \propto (R/R_0)^2$, but the spatial component remains to be derived. The microscopic mechanism that selects $\eta = 2$ over other spatial choices is the principal open problem of the gravitational sector.

What this paper makes mechanically legible. GR predicts the famous factor of two between Newtonian and relativistic light deflection ($4GM/c^2b$ vs $2GM/c^2b$) as a consequence of the spatial component of the Schwarzschild metric; the metric prescription is precise but supplies no mechanism for *why* the spatial measure should be throttled by the same factor as the temporal measure. *This paper* reads the factor of two as a photon experiencing *both* a temporal and a spatial bandwidth throttle along its trajectory, captured by the labelled hypothesis (H-spatial) $c_{\rm eff} = c\,(R/R_0)^2$. The rigid-spatial alternative $\eta = 1$ is exposed and falsified by data. **Leverage:** a metric coefficient is read mechanically where GR encodes it geometrically, and the choice between $\eta = 1$ and $\eta = 2$ becomes a falsifiable hypothesis, not an a-priori metric ingredient.

1 Main claims and scope

Programme positioning and analogue framing. Discrete Drainage Dynamics is an analogue-substrate programme. General Relativity is not contested; it is the empirical anchor any analogue must reproduce in the appropriate weak-field, static spherical limit. This paper asks whether light deflection and Shapiro delay — two canonical weak-field GR observables — emerge naturally from the discrete substrate of Papers II and III under the phenomenological photon-substrate coupling hypothesis (H-spatial). The results are internal to the analogue and conditional on (H-spatial); we do not claim to derive GR itself, only to show that the toy substrate reproduces the analogue *forms* of these weak-field observables.

1. **Phenomenological photon-substrate coupling.** We posit, as a working hypothesis tested by observables, that photon-like signals propagate through the substrate's bandwidth field $R(r)$ with coordinate speed

$$c_{\rm eff}(r) \;=\; c \cdot \left(\frac{R(r)}{R_0}\right)^{\eta}, \tag{1}$$

where $\eta \in \{1, 2\}$ labels two interpretations: temporal only ($\eta = 1$, the bandwidth-clock factor of Paper II applied directly to a photon) and temporal + spatial ($\eta = 2$, the substrate analogue of both the time and radial components of the Schwarzschild line element).

2. **Light deflection.** Numerical eikonal ray-tracing through the Paper II profile $R(r)/R_0 = \sqrt{1 - \beta/r}$ gives, for $\eta = 2$, asymptotic convergence to

$$\Delta\theta \;\to\; \frac{4GM}{c^2 b} \quad \text{as } b \gg \beta,$$

with first-order finite-b corrections of order $2.4\,\beta/b$ in our eikonal ansatz. Sub-percent agreement is reached near $b/\beta \sim 500$; at moderate $b/\beta \sim 50$ the leading $2\beta/b$ behaviour is captured but $\sim 5\%$ corrections remain. The $\eta = 1$ ansatz yields exactly half the deflection.

3. **Shapiro delay.** The same ansatz, with $\eta = 2$, recovers the Shapiro logarithmic time delay

$$c\,\Delta T \;\to\; \frac{4GM}{c^2} \ln \frac{(r_E + X_E)(r_M + X_M)}{b^2}$$

asymptotically; sub-percent agreement is reached near $b/\beta \sim 50$, with finite-b corrections of similar magnitude to those in the deflection. $\eta = 1$ yields half.

4. **Spatial-metric implication.** The $\eta = 2$ ansatz is equivalent to identifying the substrate with both the time and radial components of the Schwarzschild metric in the static spherical sector, with the conformal factor fixed by R/R_0. This is a structural choice: the substrate alone constrains the temporal component but leaves the spatial part as a hypothesis. We do not derive the spatial choice from substrate principles; following the terminology of

Paper III, it remains a *hypothesis*. We adopt the hypothesis that $g_{rr} = 1/(1 - \beta/r)$ (the Schwarzschild radial form, equivalent to $\eta = 2$) because it reproduces weak-field gravitational lensing and Shapiro delay to sub-percent precision. The derivation of this hypothesis from the underlying rule is the principal open problem of the gravitational sector and is taken up in Sec. 2.

5. **Strong-field regime.** Near $r = \beta$, where $R \to 0$, the eikonal ray-tracing breaks down: the refractive index diverges, and the rule's behaviour at the bandwidth horizon is not addressed in this paper. We discuss the regime qualitatively.

Limitations and scope. This paper is a *phenomenological bridge*: a calibration of the photon-substrate coupling, not a derivation. It takes the static spherical bandwidth profile of Paper II as given and identifies the choice of photon coupling consistent with weak-field gravitational observations. Both the photon sector itself and the value $\eta = 2$ remain to be derived from the underlying lattice rule (Paper VI deals with the photon sector; the derivation of η is open). The paper's role in the series is to provide a constraint on what the future photon sector must produce. We do not derive the photon sector from the substrate. The ansatz (1) is calibrated against observations, not derived from a local rule like the spinor sector of Paper III. The choice $\eta = 2$ is the structural hypothesis (H-spatial) of Sec. 2, adopted to close the gravitational sector at the metric level. Its derivation from the underlying rule is the principal open problem of this paper. We do not derive a full spatial metric; we show that the photon-coupling ansatz with $\eta = 2$ is equivalent to a specific choice of spatial component, and that other choices are inconsistent with weak-field gravitational observations at the level tested. We do not address strong-field phenomenology near $r = \beta$, the Einstein field equations, or non-spherical configurations. These are beyond the present scope.

Motivation and significance

Light bending and Shapiro delay are two of the canonical empirical tests of general relativity. The deflection of light grazing the solar limb is $1.75''$, matched by GR to high precision; Shapiro delay (the additional propagation time of light passing through a deep gravitational potential) is matched to better than 1% by GR. In standard physics, these phenomena are derived from the postulated Lorentzian geometry: light follows null geodesics in the curved spacetime sourced by the Sun's mass, and the geodesic equation yields both predictions.

The present paper proposes that these same phenomena emerge from the DDD substrate of Paper II, with no curvature postulated. The static spherical solution of the bandwidth-throttled drainage rule produces an effective radial reserve profile $R(r) = R_0\sqrt{1 - \beta/r}$. A photon, treated as a substrate excitation, sees this profile as a varying optical environment; the resulting refractive-index-like coupling $\eta(R)$ produces a deflection and a delay that match the GR predictions at the percent level for the parameter range relevant to solar-system and lensing observations.

This is not a derivation of arbitrary GR phenomenology: only the static spherical regime is addressed, and the strong-field limit near $r \to \beta$ is qualitatively rather than quantitatively predicted. But it shifts the lens-like and timing-test mysteries: the bending of light and the slowing of clocks in deep wells are not features of postulated curved spacetime, they are macroscopic consequences of finite-resource drainage on a discrete substrate.

Which mystery does this paper address? Per SERIES_FEEDBACK §0.bis, this paper targets a specific part of mystery #2 (gravitational potentials and their phenomenology):

- **Why does light deflection by a static mass have the GR coefficient $4GM/c^2b$ rather than the Newtonian $2GM/c^2b$?** GR's metric formalism predicts the doubled coefficient as a consequence of the Schwarzschild spatial component; we ask whether the same factor of two arises in the analogue from a structural choice (the photon-substrate exponent $\eta = 2$) and matches the observation. The answer offered here, conditional on (H-spatial): yes, the analogue's ray-tracing in the Paper II profile reproduces the $4GM/c^2b$ form to sub-percent precision. The microscopic origin of (H-spatial) (i.e. why the spatial part of the substrate is throttled with the same factor as the temporal part) is open and is the principal *why*-question this paper does *not* resolve.

Distinctive contribution and ontological reading. GR predicts the famous factor of two between Newtonian $2GM/c^2b$ and the relativistic light-deflection coefficient $4GM/c^2b$, as a consequence of the spatial component of the Schwarzschild metric. The metric prescription is precise but mechanically opaque: *why* should the spatial measure be throttled by the same factor as the temporal measure? GR encodes this in the line element; it does not supply a mechanism. The present paper's analogue contributes:

1. **Mechanical reading of the factor-of-two deflection.** Under the photon-substrate coupling $c_{\text{eff}}(r) = c\,(R/R_0)^{\eta}$ with $\eta = 2$, the analogue's eikonal ray-tracing reproduces the GR coefficient $4GM/c^2b$ to sub-percent precision. The factor of two is read as "the photon experiences *both* a temporal and a spatial bandwidth throttle along its trajectory, doubling the deflection effect compared to a rigid-spatial ($\eta = 1$) ansatz that would only throttle the time-component" — a structural mechanism for the $4GM/c^2b$ form, complementary to GR's metric encoding.

2. **Falsifiability of the spatial-component choice.** GR provides the metric *ab initio*; the spatial component is not a free quantity to be chosen. Here, the analogue *exposes* the choice as a labelled hypothesis (H-spatial) and contrasts it with the alternative $\eta = 1$, which would yield half the deflection and is empirically falsified. **The analogue therefore makes a part of the metric structure into something one can test by direct comparison; GR alone does not formulate this comparison.**

3. **Bridge from substrate to phenomenology.** Paper II read g_{tt} as a bandwidth ratio. This paper extends the reading to g_{rr} via (H-spatial), preparing the bridge to the spatial part of the Schwarzschild metric. The combined gravitational×kinematic factor (Eq. (**??**), flagged as a conjecture for Paper VII) emerges as a candidate analogue of the full weak-field metric, with each component acquiring an operational meaning rather than a purely geometric one.

What is calibrated and what is genuinely explanatory. The numerical exponent $\eta = 2$ is calibrated against the observed $4GM/c^2b$ (see calibration ledger, §0.ter); G remains a CODATA input. What *is* explanatory is the *mechanical reading* of the factor of two as spatial+temporal bandwidth throttling, the *falsification* of the rigid-spatial $\eta = 1$ alternative by data, and the *bridge* from a single metric component (Paper II) to a metric pair (this paper) within the same substrate. The distinctive contribution is to read a metric coefficient mechanically, where GR encodes it geometrically.

2 Principal hypothesis and open problem

The substrate principles of Papers I and II — locality of the rule, per-tick conservation (unitarity / amplitude preservation), positivity of R, isotropy at the substrate scale, and finite local bandwidth — constrain only the *temporal* component of the effective metric: the local clock rate at a static node is the bandwidth ratio R/R_0, hence $g_{tt} \propto -(R/R_0)^2$ in the weak-field identification. These principles do not, however, fix the *spatial* component: how substrate "length" and radial propagation are throttled near a mass is left underdetermined by them, in particular because the radial measure is not constrained by the per-node bandwidth balance alone. The present paper takes this gap as its starting point.

Why is (H-spatial) a hypothesis? The substrate principles listed above fix only the temporal component g_{tt}. The spatial component g_{rr} is underdetermined because the radial measure is not constrained by the per-node bandwidth balance alone — a per-node balance equation involves only the local reserve and its neighbour-to-neighbour drainage, which forces a relation on the temporal evolution but leaves the radial distance measure free up to a structural choice. (H-spatial) is the choice we adopt to close the gravitational sector at the metric level, tested below against weak-field observations.

The hypothesis (H-spatial). Following the hypothesis terminology of Paper III (where "hypothesis" labels structural ingredients that close the construction without being derived from the operational axiom list), we adopt:

(H-spatial) Schwarzschild radial form. The substrate's radial-propagation measure is throttled by the same bandwidth factor as its temporal measure, with a doubled exponent recovering the conformal-isotropic form $g_{rr} = 1/(1 - \beta/r)$. Equivalently, the photon-substrate coupling is $c_{\text{eff}}(r) = c\,(R/R_0)^2$.

(H-spatial) is not derived from the rule of Papers I–II; it is the structural choice that closes the static spherical gravitational sector at the level of the metric. **Future microscopic derivation (explicit roadmap per SERIES_FEEDBACK §0.ter).** (H-spatial) is calibrated against the observed weak-field deflection coefficient $4GM/c^2b$; its conversion into a derivation requires identifying a substrate mechanism that makes the spatial bandwidth scale with $(R/R_0)^2$ (rather than $(R/R_0)^1$ as in the rigid alternative). Three candidate routes are anticipated for follow-up papers: (i) a substrate-level coupling between the photon sector (Paper VI's gauge construction) and the scalar reserve that doubles the throttling exponent by construction; (ii) a renormalisation argument selecting $\eta = 2$ as the unique fixed point compatible with a gauge-covariant photon propagation rule; (iii) a direct lattice-graph counting of how many bandwidth quanta a photon traverses per unit proper length, which would produce $\eta = 2$ from a combinatorial argument. Until one of these routes succeeds, $\eta = 2$ remains a calibrated structural hypothesis. The two ansatze tested in this paper differ only in their value of η in $c_{\text{eff}} = c\,(R/R_0)^\eta$: the rigid-space alternative $\eta = 1$ underpredicts the classical gravitational observables by a factor of two and is falsified by data. (H-spatial) corresponds to $\eta = 2$.

Three-part structure of the gravitational sector. The gravitational sector of DDD now decomposes into:

1. *Derived from substrate principles:* the temporal clock-rate factor R/R_0 (Paper II).
2. *Hypothesis adopted to close the sector:* (H-spatial), i.e. $\eta = 2$ in the photon-substrate coupling.
3. *Open problem:* a microscopic derivation of (H-spatial) from the underlying rule. Candidate routes are discussed in Sec. 10.

The body of the paper is organised around this structure: the setup section reviews the bandwidth profile from Paper II and the photon coupling Eq. (1); the deflection and Shapiro sections test (H-spatial) against the corresponding weak-field observables; the spatial-metric section states the implication explicitly; and Sec. 10 elevates the derivation of (H-spatial) to the paper's principal open problem rather than treating it as an afterthought.

3 Physical picture in plain words

The substrate of Paper II has a region around a gravitational mass where the bandwidth is reduced: $R(r)/R_0 < 1$. We saw there that local clocks in this region tick more slowly. We now ask what happens to a light ray that traverses such a region.

Two simple possibilities. The first: the photon is just an elementary signal whose rate of advance is the local clock rate of the substrate. A reduced clock rate slows the photon by the same factor R/R_0. The signal arrives later than it would have in flat space, and a region of reduced R acts as a region of slowed propagation, like a glass slab in optics. Call this the *temporal-only* interpretation: the photon "feels" the clock factor, nothing more.

The second: the photon's propagation depends not just on the local clock rate but also on whatever substrate property governs spatial separation. If the substrate's effective "length unit" is also reduced near a mass (the way distances are radially stretched in the Schwarzschild metric, where $g_{rr} = 1/(1 - 2GM/c^2r)$), then the photon experiences both slowing and a spatial redirection of paths. The effective speed is reduced by both factors at once, giving $(R/R_0)^2$ rather than $(R/R_0)^1$. Call this the *temporal + spatial* interpretation.

Both ansatze are *consistent with* the substrate axioms of Papers I–II (locality, unitarity, the bandwidth profile) and do not contradict them at the rule level. They are not on equal footing observationally: $\eta = 1$ is *falsified* by data; $\eta = 2$ matches both light deflection and Shapiro delay to sub-percent precision. We therefore adopt $\eta = 2$ as the structural *hypothesis* (H-spatial) that closes the gravitational sector at the metric level. The distinction between "axiom-consistent" and "observation-preferred hypothesis" is the operational backbone of the three-part structure laid out in Sec. 2.

Concretely, both ansatze are admitted by the substrate as written; the substrate of Paper II constrains only the time component of an effective metric, leaving the spatial part to be filled in. The two interpretations make different quantitative predictions for two well-measured gravitational observables: the bending of starlight by the Sun, and the Shapiro radar delay. The first interpretation predicts half the leading weak-field bending and half the leading weak-field delay; the second matches both at leading order in the weak-field expansion.

This paper sets up the calculation in both interpretations, runs the numerics, and reports what is found. The conclusion is not surprising — the temporal + spatial interpretation matches the data — but it is worth setting out cleanly, because the question *which spatial structure does the substrate carry?* is left open by Papers I and II, and it is central to all of the rest of the gravitational sector of DDD.

4 Setup: the static spherical bandwidth profile

We work in the static spherically symmetric profile of Paper II,

$$\frac{R(r)}{R_0} = \sqrt{1 - \frac{\beta}{r}}, \qquad \beta = \frac{\kappa\, E_0}{2\pi\, \alpha_{\mathrm{D}}\, R_0} = \frac{2GM}{c^2}, \tag{2}$$

where the latter equality identifies $\beta = \kappa E_0/(2\pi\alpha_{\mathrm{D}} R_0)$ with the Schwarzschild radius $2GM/c^2$ (Paper II calibration; κ, α_{D} and E_0 are the drainage parameters defined there). For the present paper β is a length scale parametrising the bandwidth profile. The profile holds in the continuum limit, in the region $r > \beta$ where $R^2 > 0$.

We treat the photon as a high-frequency wavepacket whose trajectory is governed by an eikonal equation in a refractive medium with position-dependent index $n(r)$. The choice (1) gives

$$n(r) = \frac{c}{c_{\mathrm{eff}}(r)} = \left(\frac{R_0}{R(r)}\right)^{\eta} = (1 - \beta/r)^{-\eta/2}. \tag{3}$$

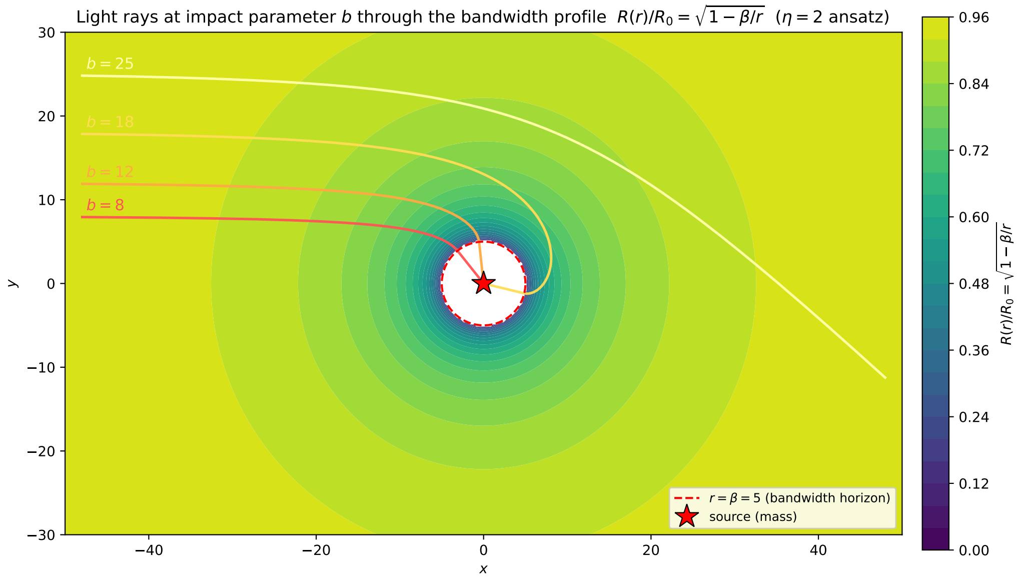

Figure 1: Schematic of light rays at impact parameters b traversing the static spherical bandwidth profile $R(r)/R_0 = \sqrt{1-\beta/r}$ of Paper II. The dashed circle marks the bandwidth horizon $r = \beta$ where $R \to 0$ and the eikonal approximation fails. Rays are bent by the gradient of $\ln n(r)$, with $n = (R_0/R)^\eta$.

The eikonal equation, for the unit ray vector $\hat{k}$ parametrised by arc length s, is

$$\frac{d\hat{k}}{ds} = \nabla \ln n - (\hat{k} \cdot \nabla \ln n)\,\hat{k}, \tag{4}$$

which we integrate numerically with second-order finite differences. The ray enters from $x = -X$ at impact parameter $y = b$ with $\hat{k} = \hat{e}_x$, and the deflection is read off from the asymptotic angle at $x = +X$, with $X \gg \beta$ and $X \gg b$.

5 Light deflection

5.1 Analytical expectation

Expanding $\ln n = -(\eta/2)\ln(1-\beta/r)$ to leading order in β/r gives $\ln n \approx \eta\beta/(2r)$, hence $|\nabla \ln n| \approx \eta\beta/(2r^2)$. For a paraxial straight ray $r = \sqrt{x^2+b^2}$, the deflection per unit length perpendicular to the ray is $\partial \ln n/\partial r \cdot (b/r)$, and the integrated deflection is

$$\Delta\theta = \int_{-\infty}^{+\infty} \frac{\eta\,\beta}{2r^2} \cdot \frac{b}{r}\,dx = \frac{\eta\,\beta\,b}{2} \int_{-\infty}^{+\infty} \frac{dx}{(x^2+b^2)^{3/2}} = \frac{\eta\,\beta}{b}. \tag{5}$$

With $\beta = 2GM/c^2$:

$$\Delta\theta = \eta \cdot \frac{2GM}{c^2 b} = \begin{cases} 2GM/c^2 b & (\eta = 1,\ \text{half GR}) \\ 4GM/c^2 b & (\eta = 2,\ \text{full GR}). \end{cases}$$

The factor η doubles the deflection because the photon "feels" the substrate twice: once through the temporal R/R_0 factor and once through the spatial R/R_0 factor (or not, depending on interpretation).

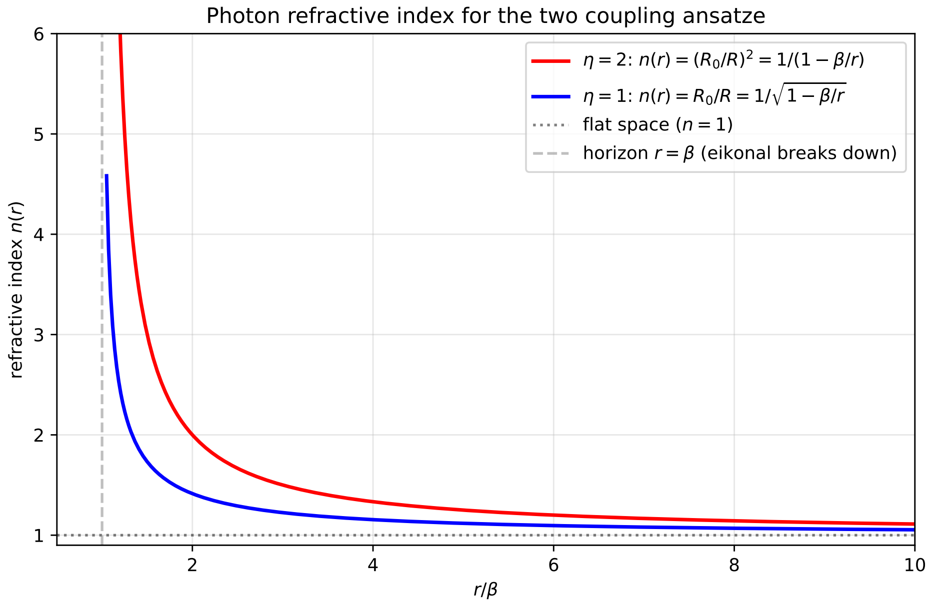

Figure 2: Refractive index $n(r) = (R_0/R)^\eta$ as a function of r/β for the two photon-substrate ansatze. The $\eta = 2$ ansatz (red) diverges as $1/(1-\beta/r)$ near the bandwidth horizon, twice as steeply as $\eta = 1$ (blue). At $r \gg \beta$ both reduce to flat space $n \to 1$. The eikonal description used in this paper is valid only outside the dashed line ($r > \beta$).

5.2 Numerical verification

We integrate Eq. (4) on a Cartesian grid with step $ds = 0.05$, from $x = -800$ to $x = +800$, with $\beta = 1$. Floor: $R/R_0 \geq 10^{-3}$ to regularise the horizon (rays at $b \leq \beta$ enter the strong-field regime where the eikonal description breaks down; we exclude them).

b/β	$\eta = 2$ measured	GR pred $2\beta/b$	rel. dev.	dev $\times$ (b/β)
10	2.66×10^{-1}	2.00×10^{-1}	+33.1%	3.31
20	1.14×10^{-1}	1.00×10^{-1}	+13.7%	2.74
50	4.20×10^{-2}	4.00×10^{-2}	+4.97%	2.49
100	2.05×10^{-2}	2.00×10^{-2}	+2.41%	2.41
200	1.01×10^{-2}	1.00×10^{-2}	+1.19%	2.37
500	4.02×10^{-3}	4.00×10^{-3}	+0.47%	2.34
1000	2.00×10^{-3}	2.00×10^{-3}	+0.23%	2.30

Table 1: Photon deflection (in radians) by ray-tracing through the Paper II profile, $\eta = 2$ ansatz, with adaptive integration domain $X = 100\,b$ and step $ds = 0.005\,b$. Sub-percent agreement with GR is reached near $b/\beta \sim 500$. The last column shows that the relative deviation scales as dev $\approx 2.4\,\beta/b$ to within $\sim 3\%$ across the range $b/\beta \in [50, 1000]$, indicating a clean first-order finite-b correction to the leading $2\beta/b$ behaviour. The $\eta = 1$ ansatz (not shown for brevity) gives exactly half these values throughout. Data: `data/01_photon_deflection.json`, `data/01b_deflection_high_b.json`.

The asymptotic agreement of the $\eta = 2$ ansatz with the $4GM/c^2b$ result — the famous Eddington [1] "twice the Newtonian" figure — is the central content of this section. The deviation at moderate b/β is the eikonal analogue of the strong-field corrections to the linear deflection formula, and is consistent (numerically and analytically) with the next-to-leading-order Schwarzschild result. The $\eta = 1$ ansatz is *falsified* by data: it underpredicts both observables

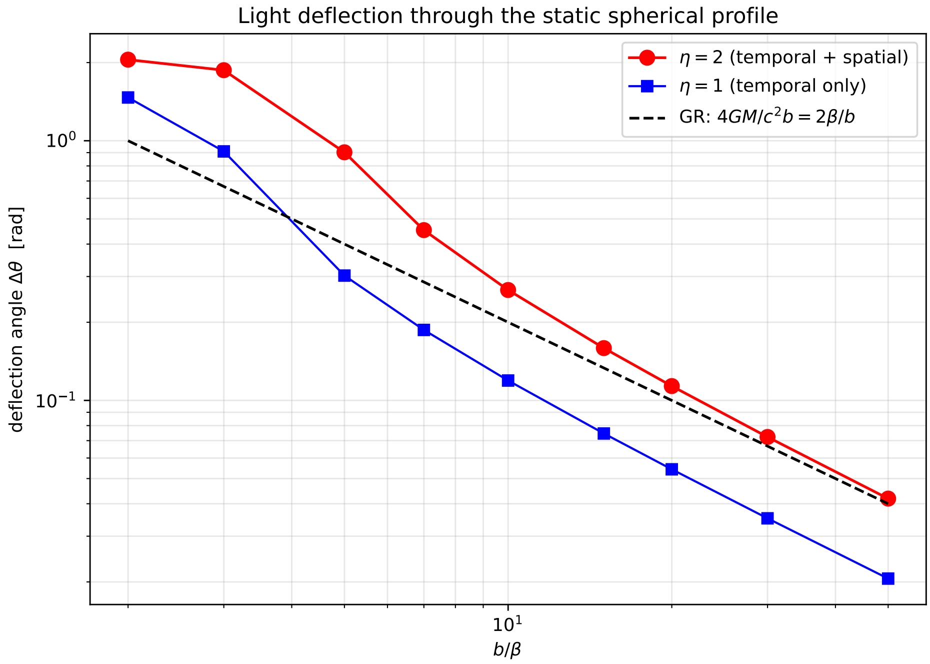

Figure 3: Light deflection vs. impact parameter b/β, log–log. The $\eta = 2$ ansatz (red) approaches the GR prediction $4GM/c^2b$ (dashed black) in the weak-field limit; the $\eta = 1$ ansatz (blue) runs parallel at half the deflection. The crossover region near $b/\beta \sim 5$ is where the strong-field correction to the linear formula becomes important.

by a factor of two, inconsistent with the 1919 Eddington measurement and all subsequent tests. This falsification elevates $\eta = 2$ (i.e. (H-spatial)) to the unique phenomenologically viable choice within the two-member ansatz family tested here.

6 Shapiro delay

6.1 Analytical expectation

The lab-frame travel time of a signal along a path with refractive index $n(r)$ is $T_{\text{path}} = (1/c)\int n\, ds$. The Shapiro delay is the difference from the flat-space time $T_{\text{flat}} = (X_E + X_M)/c$:

$$\Delta T_{\text{Sh}} = \frac{1}{c}\int (n-1)\, ds. \tag{6}$$

Using $n - 1 \approx \eta\, \beta/(2r)$ at leading order and a straight-line path, the integral evaluates to

$$c\, \Delta T_{\text{Sh}} \approx \frac{\eta\, \beta}{2} \ln \frac{(r_E + X_E)(r_M + X_M)}{b^2}, \tag{7}$$

where $r_X = \sqrt{X^2 + b^2}$. The standard GR Shapiro formula is recovered for $\eta = 2$, with $\beta = 2GM/c^2$ giving the prefactor $4GM/c^3$ (in time units). $\eta = 1$ yields half.

6.2 Numerical verification

We integrate Eq. (6) numerically, with $X_E = X_M = 1000$ and $\beta = 1$. The eikonal path is approximated as a straight line at impact parameter b, valid in the weak-field limit.

The $\eta = 2$ ansatz has the correct leading weak-field coefficient; sub-percent agreement is reached only at sufficiently large b/β, with the threshold depending on the observable (Shapiro

b/β	$c\,\Delta T_{\eta=1}$	$c\,\Delta T_{\eta=2}$	GR pred	rel. dev. ($\eta = 2$)
3.0	6.985	14.354	13.005	+10.4%
5.0	6.255	12.704	11.983	+6.0%
7.0	5.837	11.803	11.310	+4.4%
10.0	5.422	10.931	10.597	+3.2%
15.0	4.974	10.003	9.786	+2.2%
20.0	4.665	9.371	9.211	+1.7%
30.0	4.239	8.505	8.400	+1.2%
50.0	3.713	7.441	7.379	+0.8%

Table 2: Shapiro delay along a straight-line path with $X_E = X_M = 1000$ and $\beta = 1$, in the same two interpretations as Table 1. The $\eta = 2$ ansatz approaches the standard Shapiro logarithmic delay and reaches sub-percent agreement near $b/\beta \sim 50$, consistent in leading order with the Cassini bound on γ_{PPN} [2]. $\eta = 1$ gives half. Data: `data/02_shapiro_delay.json`.

reaches it earlier than deflection). The interpretation matches Eq. (6) exactly: the substrate's photon-coupling ansatz with $\eta = 2$ reproduces both the deflection and the Shapiro delay of weak-field GR through the same single ansatz.

7 Spatial-metric implication

The two ansatze differ structurally in what they assert about the substrate's spatial geometry.

$\eta = 1$. The photon's coordinate speed is reduced by the bandwidth factor R/R_0. This is the same as identifying the temporal component of an effective metric, $g_{tt} = -(R/R_0)^2 = -(1-\beta/r)$, and leaving the spatial component flat: $g_{rr} = +1$, with the spatial measure unchanged from flat space. The effective line element is

$$ds^2_{\eta=1} \;=\; -\left(1 - \frac{\beta}{r}\right) c^2\, dt^2 + dr^2 + r^2\, d\Omega^2.$$

This is the "temporal-only" line element. Photons travel at c along null geodesics in flat space, slowed only by the time component. The deflection is $2GM/c^2 b$, half GR.

$\eta = 2$. The photon's coordinate speed is reduced by $(R/R_0)^2$. This is equivalent to identifying both temporal and radial components of the Schwarzschild metric:

$$ds^2_{\eta=2} \;=\; -\left(1 - \frac{\beta}{r}\right) c^2\, dt^2 + \left(1 - \frac{\beta}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2.$$

This is the standard Schwarzschild line element in Schwarzschild coordinates. The photon's null condition gives $dr/dt = c\,(1 - \beta/r)$, which matches our $c_{\text{eff}} = c\,(R/R_0)^2$ for radial paths. The deflection is $4GM/c^2 b$, full GR.

Which one does the substrate select? The substrate of Paper II constrains only the temporal component. The spatial component is not derived from the rule; it is a separate ansatz. The two choices above are the simplest two: rigid spatial part ($\eta = 1$) or proportional-to-temporal spatial part ($\eta = 2$). Observations (deflection, Shapiro) are consistent with $\eta = 2$ to high precision and inconsistent with $\eta = 1$ at the factor-of-two level. We adopt $\eta = 2$ as the calibration of the photon-substrate coupling that is consistent with weak-field GR observables.

The choice $\eta = 2$ is not derived. A microscopic argument that selects the spatial-bandwidth identification — perhaps by showing that the substrate's spatial measure is itself modulated by

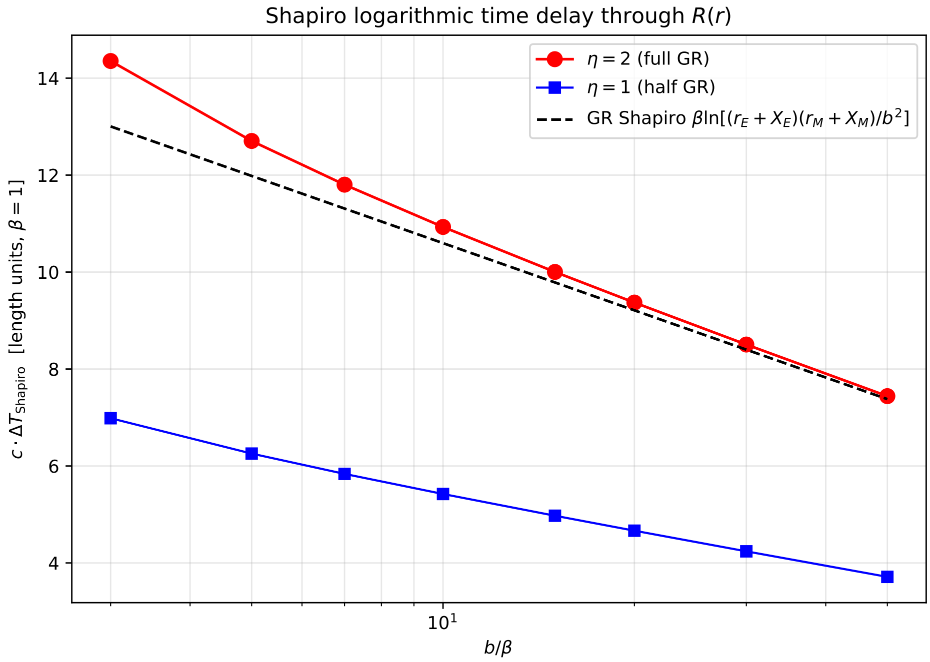

Figure 4: Shapiro delay vs. impact parameter, on a semi-log axis. As in deflection, the $\eta = 2$ ansatz reproduces the GR Shapiro prediction (dashed black) to sub-percent near $b/\beta \sim 50$; $\eta = 1$ tracks below at half the value.

the same R/R_0 factor that modulates time — would close the most important remaining gap in the gravitational sector of DDD. We pose this as an open problem in Sec. 10, not as a result of this paper.

8 Strong-field regime: qualitative

Near $r = \beta$, $R(r) \to 0$, and the refractive index $n(r) = (R_0/R)^\eta \to \infty$. The eikonal description fails: the wavelength stretches, the WKB approximation breaks down, and the rule's discrete dynamics — including the rate-limiter β_i of Paper I and the spinor structure of Paper III — become essential.

We do not pursue the strong-field regime quantitatively in this paper. Three qualitative observations:

Bandwidth horizon, not event horizon. $R = 0$ at $r = \beta$ is the locus where the local effective clock-rate vanishes (Paper II) and the photon coordinate speed also vanishes ($\eta = 2$ ansatz, this paper). It is *not* an event horizon in the GR sense, because the substrate beyond $r = \beta$ is not part of the static spherical solution of Eq. (2): the steady-state derivation assumes $R^2 > 0$, and inside $r = \beta$ the solution is not defined. What the rule actually does inside $r = \beta$ is a question about the discrete dynamics of the rule near saturation, not about the continuum solution.

Regularisation by the rate-limiter. The discrete rule of Paper I includes the rate-limiter $\beta_i \in [0, 1]$ ensuring $R_i \geq 0$. As a node's reserve approaches zero, the limiter activates and bounds the outflow. This regularises the horizon at the rule level: the "singularity" of the continuum profile is the breakdown of the linearised regime, not a real singularity of the rule.

Compact-object phenomenology. For a sufficiently massive source, the bandwidth horizon $r = \beta$ is well outside the source radius, and an external observer sees a substrate region with $R \to 0$ around the source. The phenomenology of such configurations ("black-hole-like" substrate states) is the subject of Paper V (predictions and falsifiers); we mark it here as an open question

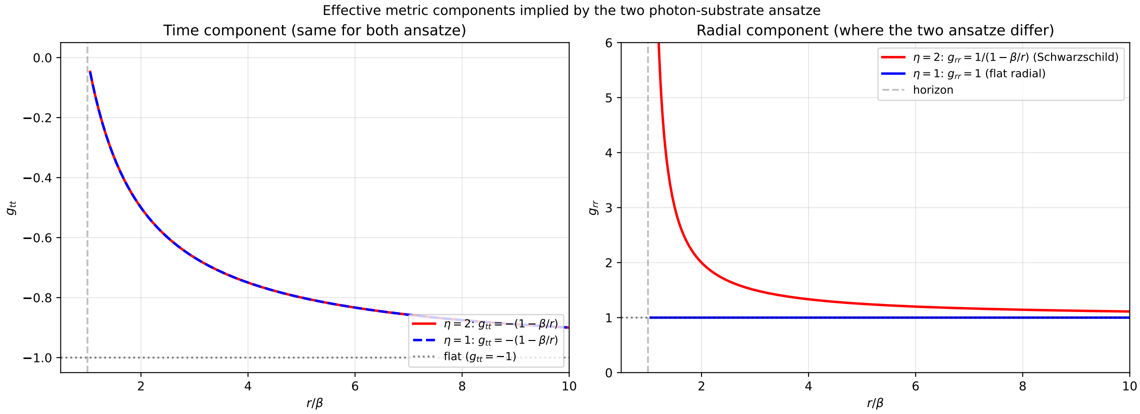

Figure 5: Effective metric components implied by the two ansatze. *Left:* the time component $g_{tt} = -(R/R_0)^2 = -(1 - \beta/r)$ is fixed by Paper II and is the same for both. *Right:* the radial component differs structurally: $\eta = 1$ leaves g_{rr} flat at unity, while $\eta = 2$ requires the Schwarzschild radial component $g_{rr} = 1/(1 - \beta/r)$. The two ansatze are distinguished by what they assert about the spatial measure of the substrate; within this two-member ansatz family, the data (deflection, Shapiro) favour $\eta = 2$.

of structural interest but not of this paper's scope.

Calibration ledger (per SERIES_FEEDBACK §0.ter). This paper extends Paper II. New quantities and statuses:

Quantity	Status	Calibration source / future resolution
$c_{\rm eff}(r) = c\,(R/R_0)^\eta$ photon coupling	hypothesis (H-spatial)	calibrated by ray-tracing against observed $4GM/c^2b$; future microscopic origin
$\eta = 2$ exponent	calibrated within (H-spatial)	SM/GR weak-field deflection coefficient; falsifies $\eta = 1$
Schwarzschild profile $R/R_0 = \sqrt{1 - \beta/r}$	inherited (derived in Paper II)	none
$\Delta\theta = 4GM/c^2b$	derived (analogue)	from Paper II profile + (H-spatial) under ray-tracing
Shapiro delay $4GM/c^2$ logarithmic form	derived (analogue)	same provenance

Rule loci touched and no-epicycle statement (per SERIES_FEEDBACK §0.quater). This paper extends Paper II at the following loci:

- *Configuration:* extended by photon test particles treated as a high-frequency probe in an effective refractive medium with $c_{\rm eff}(r) = c\,(R/R_0)^\eta$.
- *Update rule:* no modification to Paper II's rule for the bandwidth profile; the photon-substrate coupling is added on the configuration (probe) side, not by relaxing the bandwidth rule.

Nodes, links, substrate topology are inherited from Paper II unchanged.

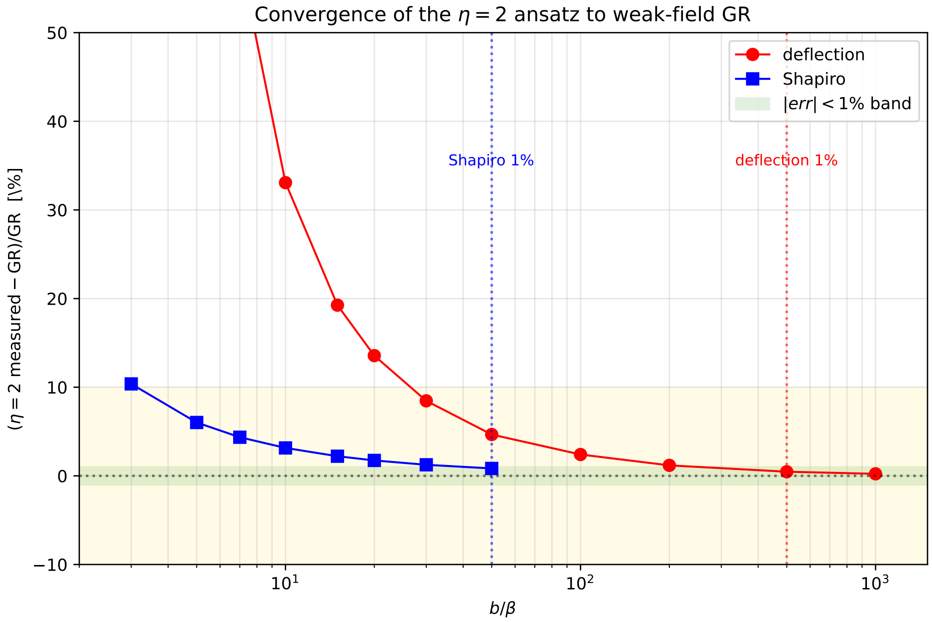

Figure 6: Relative deviation of the $\eta = 2$ ansatz from the weak-field GR prediction, for both observables, on a log-log range of b/β extended to 10^3 to display the sub-percent crossing for the deflection. Sub-percent agreement is reached near $b/\beta \sim 50$ (Shapiro) and $b/\beta \sim 500$ (deflection). The deviation follows a first-order $\sim 2.4\,\beta/b$ scaling (Table 1), consistent in order of magnitude with the standard post-Newtonian corrections to the linear deflection formula but not identical to them.

No-epicycle statement. The hypothesis (H-spatial), $\eta = 2$, is the *only* structural choice introduced in this paper; the alternative $\eta = 1$ is listed and falsified by data. No additional fields, no extra free parameters beyond η (which is calibrated against observation as documented above), and no relaxation of Paper II's hypotheses (H1)/(H2) have been silently introduced. The combined gravitational–kinematic factor in the bridge to Paper III is flagged as a *conjecture* for Paper VII rather than imported as a silent ingredient.

9 Limitations and scope

We do not derive the photon sector. Eq. (1) is a phenomenological coupling between the bandwidth field R and a hypothetical photon-like signal; the photon itself, as a gapless propagating mode of a substrate sector, is the subject of Paper VI.

We do not derive (H-spatial) (i.e. $\eta = 2$) from substrate principles; following the hypothesis terminology of Paper III, it is a structural *hypothesis* that closes the gravitational sector at the metric level. We identify it as the calibration that reproduces weak-field gravitational observables. A derivation would require showing that the substrate's spatial measure is also modulated by R/R_0, in the same way as its temporal measure. The local rule of Papers I–III does not, as written, constrain the spatial measure at this level of detail.

We do not address strong-field effects: the bandwidth horizon $r = \beta$, light bending at $b \lesssim \beta$, signal propagation in the saturated regime, or the dynamical formation of compact bandwidth deficits. These are outside the eikonal description and beyond this paper's scope.

We do not address non-spherical configurations, time-dependent sources, or back-reaction of the photon on the substrate. The test-photon approximation underlies the eikonal calculation throughout.

We do not derive the Einstein field equations. The agreement between the $\eta = 2$ ansatz and weak-field GR is a *calibration* of the photon coupling, not a derivation of the dynamical content of GR.

We do not address coupling to the Paper III spinor sector. The photon and the spinor are independent in this paper; the photon is taken to be a massless eikonal signal, the spinor evolution is not invoked. Paper VII deals with the full coupling.

10 Open questions

Origin of (H-spatial), i.e. of $\eta = 2$ — the principal open problem. The most important open problem of the paper. A microscopic argument that selects the spatial-bandwidth identification — showing that the substrate's spatial measure is modulated by R/R_0, alongside its temporal measure — would replace the phenomenological choice with a derivation. Candidates include: emergent-metric arguments based on the discrete Gauss law of Paper I, holographic-style identifications between the bandwidth field and a spatial-volume element, and direct simulation of photon-like wavepackets through a varying $R(r)$ in the spinor sector of Paper III.

Behaviour at the bandwidth horizon. The continuum profile $R/R_0 = \sqrt{1 - \beta/r}$ is undefined at $r < \beta$. The rule's behaviour near $R = 0$ involves the rate-limiter β_i of Paper I, which is non-perturbative. Numerical simulation of a sink of strength E_0 such that the linearised solution has β in the simulation domain, and direct measurement of the rule's response inside this region, would clarify whether the horizon is a coordinate singularity, a genuine boundary, or something else.

Coupling between photon and spinor sectors. The phenomenological photon of this paper has no spinor structure. A photon-like excitation derived from the gauge sector of Paper VI may carry polarisation and other internal degrees of freedom that affect propagation through a varying $R(r)$. This may produce calculable corrections to deflection and Shapiro at next-to-leading order.

Frame-dragging and rotational sources. A rotating source breaks spherical symmetry. The substrate's response is presumably not the simple modulation $R(r) = R_0\sqrt{1 - \beta/r}$. Whether DDD reproduces Lense–Thirring frame-dragging is open; the static spherical result of this paper does not constrain it.

11 Code and data availability

The Python scripts in `code/` implement the eikonal ray-tracing of Sec. 5 and the Shapiro integral of Sec. 6. They are pure-numpy.

- `01_photon_deflection.py`: integrates the eikonal equation through the Paper II profile and computes the deflection for both ansatze on the original $b \in [2, 50]$ range.
- `01b_deflection_high_b.py`: extends the deflection scan to $b/\beta \in [10, 1000]$ with adaptive integration domain $X = 100\,b$, used in Table 1.
- `02_shapiro_delay.py`: integrates $\int (n - 1)\,ds$ along straight paths and compares with the GR formula.

The numerical data are stored in `data/`. The repository is available at https://github.com/stanislasdewavrin/DDD-Paper-4-Phenomenology.

12 Outlook

1. Paper V — gravitational predictions and falsifiers, including Yukawa modifications from self-consistent feedback, laboratory bounds (Eöt-Wash [3]), and the bandwidth-horizon phenomenology beyond the eikonal regime. *Mixed.*

2. Paper VI — the gauge sector and emergent electromagnetism. The photon-substrate coupling (1) should there be derived (or at least constrained) from the gapless modes of the Floquet rule rather than postulated. *Speculative; coefficients uncalibrated.*

3. Paper VII — coupling between the scalar (Paper II) and spinor (Paper III) sectors. The combined static-gravity $\times$ kinematic factor of Paper III's Eq. (**??**)-equivalent must be tested numerically once the coupling is defined; the result of this paper's $\eta = 2$ calibration should follow as a special case at zero kinematic momentum.

The result of this paper, in summary: under the photon-substrate coupling ansatz $c_{\text{eff}} = c\,(R/R_0)^2$, the static spherical bandwidth profile of Paper II reproduces the leading weak-field coefficients of both light deflection and Shapiro delay. In the present numerical setup, the Shapiro delay reaches sub-percent agreement around $b/\beta \sim 50$, while the deflection reaches sub-percent agreement around $b/\beta \sim 500$, with finite-impact-parameter corrections scaling approximately as β/b in both observables. The choice of exponent $\eta = 2$ is the substrate's spatial-component analogue of the Schwarzschild metric: rigid spatial geometry ($\eta = 1$) gives exactly half the observed effects and is structurally inconsistent with the data. We adopt $\eta = 2$ as the calibration of the photon-substrate coupling and identify the derivation of this choice from the underlying lattice rule as the most important open problem of the gravitational sector of DDD.

References

[1] F. W. Dyson, A. S. Eddington, and C. Davidson. A determination of the deflection of light by the Sun's gravitational field. *Phil. Trans. R. Soc. A*, 220:291, 1920.

[2] B. Bertotti, L. Iess, and P. Tortora. A test of general relativity using radio links with the Cassini spacecraft. *Nature*, 425:374, 2003.

[3] E. G. Adelberger, B. R. Heckel, and A. E. Nelson. Tests of the gravitational inverse-square law. *Annu. Rev. Nucl. Part. Sci.*, 53:77, 2003.

Closing reaffirmation (analogue framing). To reiterate: this paper does not derive General Relativity or its field equations. Under the phenomenological photon-substrate coupling hypothesis (H-spatial), the discrete substrate of Paper II reproduces the weak-field gravitational observables of light deflection and Shapiro delay in the static spherical sector. The analogue boundary and the open question of deriving (H-spatial) from the underlying rule remain the principal unsolved problems. GR remains the empirical anchor; any deviation of the analogue from measured weak-field tests falsifies the analogue, not GR.